

VENKATA SAI KIRAN KUMAR V
Lead Full Stack .NET Developer
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PROFILE SUMMARY

- ❖ Results-driven Technology Leader with 18+ years of experience in full-cycle Software Development Life Cycle (SDLC), specializing in Microsoft .NET technologies, cloud-native architectures, and enterprise-scale solutions. Proven expertise in designing, architecting, and delivering scalable n-tier applications and microservices using .NET Core, ASP.NET MVC/Web API, C#.NET, VB.NET, and modern front-end frameworks like Angular (v14). Adept at leading cross-functional teams, driving best practices, and implementing secure, high-performance solutions across healthcare, financial, and loyalty domains.



CORE COMPETENCIES

- ❖ **Enterprise Architecture & Leadership:** Designed and led implementation of **microservices-based architecture** using .NET Core, CQRS, MediatR, DDD, and Clean Architecture, ensuring scalability and maintainability.
- ❖ Hands-on experience integrating LLM-based agent architectures into enterprise applications using **LangChain, AutoGen, LlamaIndex, and Semantic Kernel**.
- ❖ Designed and implemented **AI orchestration frameworks** for metadata **curation, reporting, and intelligent automation**.
- ❖ Leveraged **Azure Cognitive Services** and **Azure OpenAI** for **NLP, embeddings, and intelligent search solutions**.
- ❖ **Cloud Solutions:** Extensive experience deploying solutions on **Azure and AWS**, including **Azure Functions, Service Bus, Blob Storage, API Management, Key Vault, and Terraform-based infrastructure automation**.
- ❖ **Event-Driven Systems:** Architected **asynchronous, real-time microservices** leveraging **Azure Service Bus, AWS MSK, and distributed caching with Redis and Apache Ignite**.
- ❖ **Security & Compliance:** Implemented **RBAC, OAuth2/JWT, and Azure Key Vault integration** for secure secret management and compliance.
- ❖ **Observability & Performance:** Configured **Azure Application Insights, Log Analytics**, and centralized monitoring with **Splunk/Dynatrace** for proactive diagnostics and performance optimization.
- ❖ **Data Engineering & Reporting:** Designed **ETL pipelines to Snowflake**, built reporting frameworks using **.NET 8/9, Blazor**, and integrated **Power BI** for analytics.
- ❖ **Database Expertise:** Strong hands-on experience with **SQL Server, PostgreSQL, MySQL, and MongoDB**, including schema design, query optimization, and high-volume data processing.
- ❖ **Front-End Development:** Delivered **SPA solutions using Angular (v14), RxJS, and Nx Workspace**, ensuring modular architecture and superior UI/UX.
- ❖ **DevOps & CI/CD:** Managed **Azure DevOps pipelines, GitHub Actions, and JFrog Artifactory** for automated builds, deployments, and artifact management.
- ❖ **Automotive Integrations:** Implemented **STARBOD APIs** for secure patient data exchange and optimized **MongoDB aggregation pipelines** for real-time analytics.

- ❖ Expertise in developing and deploying the application using **Google's Vertex AI**
- ❖ Versatile Full Stack Developer with hands-on expertise in **ASP.NET MVC, .NET MAUI, and modern CI/CD practices, delivering scalable web and cross-platform mobile solutions integrated with STARBOD APIs**, Terraform automation, and enterprise observability tools like Splunk and Dynatrace.



Experience

- ❖ Working as Technical Architect with Mastercard from Feb 2024 till date
- ❖ Worked as .NET Tech Lead with Tech Mahindra from Nov 2007 till Feb 2024
- ❖ Worked as a Senior Software Engineer with Vinson Industries Pvt. Ltd., from May 2005 till October 2007
- ❖ Worked as a Software Engineer with Synergy Services Corporation from March 2003 till Apr 2005



Technical Skills

Backend	.NET Core, ASP.NET MVC/Web API, C#, VB.NET, Microservices, EF Core, Dapper, ADO.NET, .NET 8.0, .NET 9.0, Blazor, Visual Studio .NET
Frontend	Angular (14), React (basic), HTML5, CSS3, JavaScript, jQuery, Bootstrap, Material UI, =
Databases	SQL Server, PostgreSQL, MySQL, Redis, Apache Ignite, SSRS, MongoDB (basic), Oracle, PL/SQL, SQL Server DB Development, Databricks, Snowflake
Cloud	AWS S3, Lambda, API Gateway, Lambda, DynamoDB, Azure (Functions, Service Bus, Blob, APIM, AKS, ACR, App Insights, Key Vault, AD)
DevOps/Tools	Azure DevOps, GitHub Actions, Docker, Terraform (basic), YAML pipelines, SonarQube
Testing/Monitoring	XUnit, NUnit, JUnit, Postman, Swagger, Application Insights, New Relic, Jest, Cypress, Application Testing
Messaging & Queues	Azure Service Bus, Apache Kafka, RabbitMQ (basic), Event Grid
Architecture	DDD, Clean Architecture, CQRS, Event Sourcing, SOLID, Repository Pattern, REST/GraphQL, SOA, Microservices
Security	JWT, OAuth2, OpenID Connect, LDAP, Key-based Auth, Azure AD/Managed Identities
Infrastructure as Code:	Terraform (AKS, Key Vaults, CI/CD integration), Cloud Formation Templates
Reporting/BI Tools:	Power BI, Crystal Reports, SSRS
AI & LLM Integration	LangChain, AutoGen, LlamaIndex, Semantic Kernel, Azure OpenAI, Hugging Face APIs

Vector Databases	Pinecone, FAISS, Weaviate
AI DevOps	MLflow, Azure ML, CI/CD for AI workloads
Version Control	Git, GitHub, Azure Repos, TFS, Bitbucket

Projects

Client: Master Card

Jan 2025 – Till Date

Role: Technical Architect

Location: Franklin, TN (Remote)

Project: AI-Powered Metadata Governance & Reporting

Project Description

The Data Governance using Atlan project is to migrate metadata from various sources to Atlan. Atlan is data governance product which is used to provide data governance capabilities to data stewards, data curators to enrich the data assets.

Responsibilities:

- Create technical architectural documents to create a migration methodology application.
- Integrated LLM-based orchestration frameworks (LangChain, Semantic Kernel) to enrich metadata assets in Atlan.
- Designed AI-powered reporting application leveraging Azure OpenAI for summarization and insights.
- Implemented vector search using Pinecone for metadata retrieval and curation.
- Architected AI agent workflows for data stewards and curators to automate metadata enrichment.
- Collaborated with cross-functional teams in Agile sprints, ensuring AI features aligned with governance requirements.
- Created a new reporting application to consume Atlan metadata APIs to generate various reports.
- Implemented business, persistence, database, and web service layers using ASP.NET MVC, applying the Repository Pattern for better separation of concerns and testability.
- Set up monitoring and alerting with AWS CloudWatch for APIs, databases, and infrastructure health.
- Created architecture for MDLH (Metadata Lake House) in Atlan
- Built and deployed a web application leveraging Google Vertex AI for model training, prediction, and automated data processing.
- Optimize PostgreSQL performance through schema design, query tuning, and replication strategies.
- Worked with business users and created business rules data steward, data consumers, data curators, platform users
- Ensure data security and integrity across S3, PostgreSQL, and API layers with encryption and access controls.

Client: Master Card
Role: Technical Architect
Location: Franklin, TN (Remote)
Project: Elite Data Inventory

Feb 2024 – Dec 2024

Project Description

The Elite Data Inventory project is to create metadata in the Informatica Data CatLog for various Master Card data assets and curate the metadata. The project includes a Curation UI Web application and exposed wrapper services to other applications to update metadata.

Responsibilities:

- Create technical architectural documents to create a Curation web application
- Developed and maintained customer-facing applications with .NET Framework, ASP.NET MVC, and Angular
- Designed and optimized Oracle database and enhancing application responsiveness
- Create wrapper services to consume Informatica EDC and expose them to third-party applications
- Created a database and wrote PL/SQL procedures to maintain asset updates in the database
- Created Power BI reports for asset analytics

Client: Nissan North America
Role: Solution Architect
Location: Franklin, TN
Project: DBS Modernization

Feb 2022 – Feb 2024

Project Description

Dealer Business System (DBS) is running in Legacy technology as a monolithic application. The application has 4 modules (Sales, Services, Parts, Admin). This project is used to containerize the applications and deploy them in AWS. This project started with Services module modernization with repair orders, warranty claims, service appointments, and various other transactions. Repair order transactions are huge (2000 per minute) from multiple sources. Have created streaming services to process the data. Have used domain-driven design architectural patterns to design the application.

Responsibilities:

- Created cross-functional diagrams to represent the proposed technical flow.
- Lead end-to-end development for enterprise applications using **Angular, .NET Core, and AWS Cloud services.**
- Collaborate with cross-functional teams to deliver microservices-based architecture, enhancing scalability and maintainability.
- Collaborating with product owners and managers to understand business project requirements.
- Collaborating with Enterprise Architect team to deliver Solution Concepts, and estimations for the new requirements.
- Analyse business requirements and create a Logical Data model.
- Work with enterprise architects to create the solution and present them for **ERB review.**

- Integrated advanced authentication protocols, enhancing security measures and improving compliance with industry standards.
- Lead the team to deliver the project successfully.
- Worked in performance tuning.
- Used **.Net Web Api to create Microservices, dockize, and deploy** them into AWS ECS.
- Development of web, enterprise components, and messaging components using AWS MSK.
- Implemented containerized applications using **Docker** and deployed to **Kubernetes**, enabling efficient orchestration and load balancing.
- Implemented Restful Web Services to retrieve data from the client side using Micro Services architecture.
- Created domain-driven design (**DDD**) architectural patterns to identify functional domains to create Microservices.
- Built the S3 buckets and configured the event bridge on S3 buckets to trigger the Lambda functions.
- Implemented Kafka cluster and created procedures to create RO (Repair Order) topics in brokers. Created consumers to listen to the topic, retrieve data, and send them to downstream systems.
- Implemented CQRS design pattern to accept warranty claims using Kafka and write them into the reporting database.
- Implement circuit breaker patterns using the resilience4j framework to send the data to downstream systems.
- Managed the IAM rules to grant permissions to the S3 bucket to trigger the lambda.
- Used the AWS Event Bridge SDK to run the jobs to load the inbound files.
- Configured the alarms in the CloudWatch service for monitoring the server's performance.
- Extensively used SQL, PG/SQL in constructing views, indexes, stored procedures, triggers, cursors, functions, and relational database models.

Client: Nissan North America

Jan 2021 – Jan 2022

Role: Technical Lead

Location: Franklin, TN

Project: DBS Migration to AWS

Project Description

Dealer Business System (DBS) runs legacy technology as a monolithic application using the Oracle database. The application is running on on-prem. This project will lift shift and migrate the application to AWS using PostgreSQL.

Responsibilities:

- Involved in gathering project requirements.
- Implemented business applications using C# and ASP.NET, focusing on modularity and reusability.
- Rewrite the procedures from Oracle to PostgreSQL
- Led the migration of legacy systems to .NET Core and Angular, reducing maintenance costs by 40%
- Modify the SQLs from Oracle to PostgreSQL
- Work on the application code to support PostgreSQL.
- Move the code to AWS and deploy them into EC2.
- Create global temporary tables.

- Mentor junior developers, conduct code reviews, and implement the best software development and deployment practices.
- Extensively used SQL, PG/SQL in constructing views, indexes, stored procedures, triggers, cursors, functions, and relational database models.

Client: Nissan North America

Apr 2020 – Jan 2021

Role: Technical Lead

Location: Visakhapatnam, India

Project: MCM (Marketing Content Management)

Project Description:

- MCM (Marketing Content Management) is a replacement for Autodata GPAS data. Currently, Autodata is collecting marketing descriptions of Nissan and Infiniti vehicles from various sources like CMM, VPOG, Accessories, etc., maintaining their database and providing data to Global PACE and Regional (NNA, NCI, etc.) applications. MCM application is a full-stack application, having its user interface and Nissan in-house database.
- Users can enter Nissan and Infiniti marketing descriptions and provide the required model data information to global PACE. Currently, MCM is designed to support the maintenance of NCI marketing data. We can enhance this application for the maintenance of other regional markets as well.
- Contribution: Architected and delivered cloud-native solutions leveraging AWS Lambda Functions, Logic Apps, API Gateway, Secrets, and Managed Identities, ensuring scalability, security, and cost optimization.
- Led a team of 14 developers in designing and deploying event-driven microservices using Kafka, enabling real-time loyalty transaction processing.
- Implemented CI/CD pipelines using AWS DevOps (YAML) and GitHub Actions, supporting multiple daily deployments and automated testing for production-ready releases.
- Automated infrastructure provisioning with Terraform, including EKS clusters, networking, and Secrets, integrated into DevOps pipelines for seamless deployments.
- Designed secure API governance using AWS API Management, implementing policies for JWT validation, OAuth2, CORS, and Managed Identity authentication.
- Developed React-based admin dashboards, integrating REST and GraphQL APIs, and implemented Jest and Cypress for unit and end-to-end testing.
- Optimized React application performance through lazy loading, code splitting, OnPush change detection, and Nx Workspace modularization for efficient builds.
- Implemented caching strategies using Redis and in-memory caching, reducing latency and improving API response times for high-volume transactions.
- Architected .NET Core microservices with Clean Architecture, integrating MySQL, SMTP services, Service Bus, and JWT-based authentication for secure communication.
- Enhanced database performance by optimizing SQL schemas, designing efficient queries, and implementing indexing strategies for large-scale transactional data.
- Established Agile practices using Jira, driving sprint planning, backlog refinement, and retrospectives for continuous delivery.
- Collaborated with QA, DevOps, and UI teams to ensure application scalability, security, and compliance across all SDLC phases.

- Authored technical documentation detailing system architecture, workflows, and API contracts, supporting knowledge transfer and onboarding.
- Configured Azure Application Insights for proactive monitoring, diagnostics, and performance tuning in production environments.
- Mentored developers on best practices in React, .NET Core, and DevOps, fostering a culture of quality, automation, and innovation.

Client: Nissan North America

Apr 2018 – Mar 2020

Role: Senior Developer

Location: Visakhapatnam, India

Project: NCI PACE Implementation

Project Description

NCI PACE is a project to publish global marketing information to the PACE platform for the NCI region. The application connects to multiple systems and updates PACE data like inventory, offers, and dealer locators. The application receives leads from PACE stores them and sends them to Shift Digital. Expose the APIs to the Global PACE platform to provide the data.

Responsibilities:

- Architected and developed ASP.NET Core Web APIs using Microservices architecture with EF Core (Code-First), ensuring seamless inter-service communication via HttpClientFactory and Context Accessors.
- Led a cross-functional team of 9 developers to build a high-performance Loyalty Management Platform on .NET Core and Azure, focusing on promotion and rewards modules for enterprise-scale deployments.
- Implemented RESTful services with standardized JSON responses, ensuring interoperability and reliable client-server communication.
- Optimized Entity Framework performance using Compiled Queries, Multiple Contexts, Explicit Loading, and batch-saving strategies; leveraged LINQ and Lambda extensively for complex queries.
- Designed and implemented distributed caching using Apache Ignite, including console applications for cache population, reducing DB load and improving response times.
- Enhanced application security by implementing JWT authentication and OAuth2 authorization across frontend and backend systems.
- Adopted Test-Driven Development (TDD) and integrated SonarQube for static code analysis, ensuring maintainability, security, and high-quality code.
- Developed automated unit and integration tests:
 - For .NET Core APIs using XUnit
 - For React components using Jest and React Testing Library
- Created SQL scripts, stored procedures, and console utilities for production data updates, transaction logging, and error handling.
- Integrated messaging systems using SMTP (Gmail, Outlook) and Twilio, and implemented Kafka for asynchronous communication and queuing.
- Developed and optimized Kafka producers/consumers in .NET and Python, configuring Topics, Partitions, and Replication Factors for scalability and fault tolerance.

- Integrated TDD and SonarQube into CI/CD pipelines, enabling automated testing and early bug detection, improving release stability and coverage.
- Worked with AWS storage services such as S3 and EFS, handling structured and unstructured data for enterprise workloads.
- Collaborated with DevOps teams to enhance CI/CD pipelines using Bitbucket and Azure DevOps, ensuring smooth deployments.
- Monitored application performance using New Relic, identifying bottlenecks and implementing optimizations for improved responsiveness.

Client: Nissan North America
Role: Technical Lead
Location: Franklin, TN
Project: Dealer Portal Solution

Jun 2017 – Mar 2018

Project Description

Dealer Portal is an application that dealers use to navigate to all other applications. The internal users would publish the content and add the links using AEM author. The publisher module would publish the content to Dealer Portal. Spring batch applications were developed to sync the data from host systems to Dealer Portal.

Responsibilities:

- Created and tested .NET applications using Visual Studio, C#, and SQL Server.
- Developed Power BI dashboards for operational reporting and business insights.
- Designed and developed interactive Power BI dashboards to visualize KPIs, operational metrics, and business trends.
- Involved in gathering project requirements.
- Developed Spring Batch applications to sync the data from AD, mainframe.

Client: Nissan North America
Role: Senior Developer
Location: Smyrna, TN
Project: Slotting and Sequencing Implementation
Project Description:

Jan 2016 – Jun 2017

This system is used to produce slotting and sequencing results for manufacturing plan vehicle production. The constraints and orders would be sent to Optessa MLS and the MLS slot / sequence the orders. The system would receive the schedule update the OCFs and send the schedule vehicle chain to plant floor systems.

Responsibilities:

- Designed and developed web pages using ASP.NET MVC, C#.NET, HTML, JavaScript, AJAX, jQuery, CSS, and server controls.
- Used Angular as the frontend and Java as the backend.

- Produce technical specifications for complex core business and technical architecture components and systems.
- Code review, performance testing, and Improving coordination among the teams
- Worked on the Complete SDLC life cycle using Agile Methodologies (Scrum).
- Transitioning the requirements to offshore and leading the team to complete the project before the milestones successfully.
- Responsible for architecture, design, development, and deployment from start to finish.
- Involved in requirement gathering, analysis, designing, and creating HLD, and LLD documents.
- Wrote SQL queries, SQL packages, and stored procedures for database persistence.

Client: Nissan North America

Nov 2014 – Dec 2016

Role: Senior Business Analyst

Location: Smyrna, TN

Project: Decherd Forging and Casting

Project Description:

This system is used to produce and track forging crankshafts for in-house usage and external customers. Nissan's Decherd facility receives bar stock from suppliers and produces in-process and forging parts for its machining production. This project is designed to order and maintain the production and inventory at various levels, track the part's traceability, and export to other facilities.

Responsibilities:

- Work with business users in creating a project specification document
- Creating DFDs, UML diagrams
- Work with stakeholders to get sign-off on PSD
- Work on creating technical requirement documents
- Explain the requirements to the development team
- Conducting functional and technical reviews
- Work with stakeholders and drive the project till implementation
- Responsible for architecture, design, development, and deployment from start to finish.
- Involved in requirement gathering, analysis, designing, and creating HLD, and LLD documents.

Client: Nissan North America

Nov 2013 – Oct 2014

Role: Senior Business Analyst

Location: Chennai, India

Project: Stamping Plant Transfer

Project Description:

The system is used to move the stamping parts from one facility to another facility. The users will raise the requests against forecast production on the owning facility. Then the customer facility will create the schedule to produce the stamping parts and maintain the inventory. Against the delivery schedule, the owning facility creates the shipment to the receiving facility and ships the parts. Once the parts are shipped the shipment gets closed automatically.

Responsibilities:

- Involved in gathering project requirements.
- Implemented Java Spring Batch applications with multi-threading
- Worked on writing stored procedures in Oracle.

Client: Nissan North America**Feb 2011 – Oct 2013****Role: Senior Business Analyst****Location: Smyrna, TN****Project: DRAGON****Project Description:**

DRAGON has several modules in web application, windows application, console application, oracle packages, and web services. It is used to accept the orders (monthly / weekly / daily) from 114 countries and process them through NNA System and prepare master schedule and publish them. This is one of the very critical projects for NNA where in it has direct access to their head quarter systems (NML).

Responsibilities:

- Discussing with Business analysts to understand the requirements
- Produce technical specifications for complex core business and technical architecture components and systems.
- Responsible for architecture, design, development and deployment from start to finish.
- Developed .Net 4.0 Multithreaded Windows services to batch process different report requests.
- Developed WCF services which is called from report processor object for formula interpretation.
- Source Safe management.
- Created windows services to batch process
- Creating requirement understanding documents, data flow diagrams, detail design documents
- Transitioning to the offshore team and track the deliverables for all the phases
- Given solutions for better interface processing
- Responsible for all the deliverables, quality, client escalations

Client: Nissan North America**Nov 2007 – Feb 2011****Role: Senior Technical Lead****Location: Chennai, India****Project: Multiple Dev Projects and support****Project Description:**

The maintenance/production support process to Nissan & Infiniti runs through 6 business domains viz. 1) Corporate Functions 2) Sales & Marketing 3) Manufacturing 4) Auto Finance 5) Engineering and 6) NCI (Nissan Canada) through 12 packages.

Responsibilities:

- Worked on various minor enhancements using WCF, .Net, ASP, Visual Basic 6.0

- Worked on Data Access Layer modules of SCM (Supply Chain Management System) that fetch data from DB2 using Oracle PL/SQL and .Net objects.
- Analysis, design and development of Application based on business requirements.
- Responsible of system break down time as per SLAs.

May'05 - Oct'07 with Vinson Industries Pvt. Ltd. as Senior Software Engineer

- Interacted with business stakeholders and end-users to comprehend the business processes, which led to the development of C#.Net Web Services for transmitting warehouse information to the finance department.
- Played a pivotal role in crafting diverse systems, including the Student Maintenance System, Inventory Maintenance System, and Office Automation System.
- Developed and implemented a C#.Net-based Web Services Service to optimize the transmission of warehouse information to finance, resulting in a 95% improvement in data accuracy and operational efficiency.

Mar'03 -Apr'05 with Synergy Services Corporation as Software Engineer

- Contributed to the development of various systems such as the Student Maintenance System, Inventory Maintenance System, Office Automation System, and more.



EDUCATION

- 2010: Master of Computer Application - [Alagappa University], Chennai
- 1998: Bachelor of Commerce – Andhra University, Visakhapatnam



PERSONAL DETAILS

- Address: 17107, Crowne Brook Circle, Franklin, TN - 37067
- Languages: English, Hindi, Telugu, and Tamil.